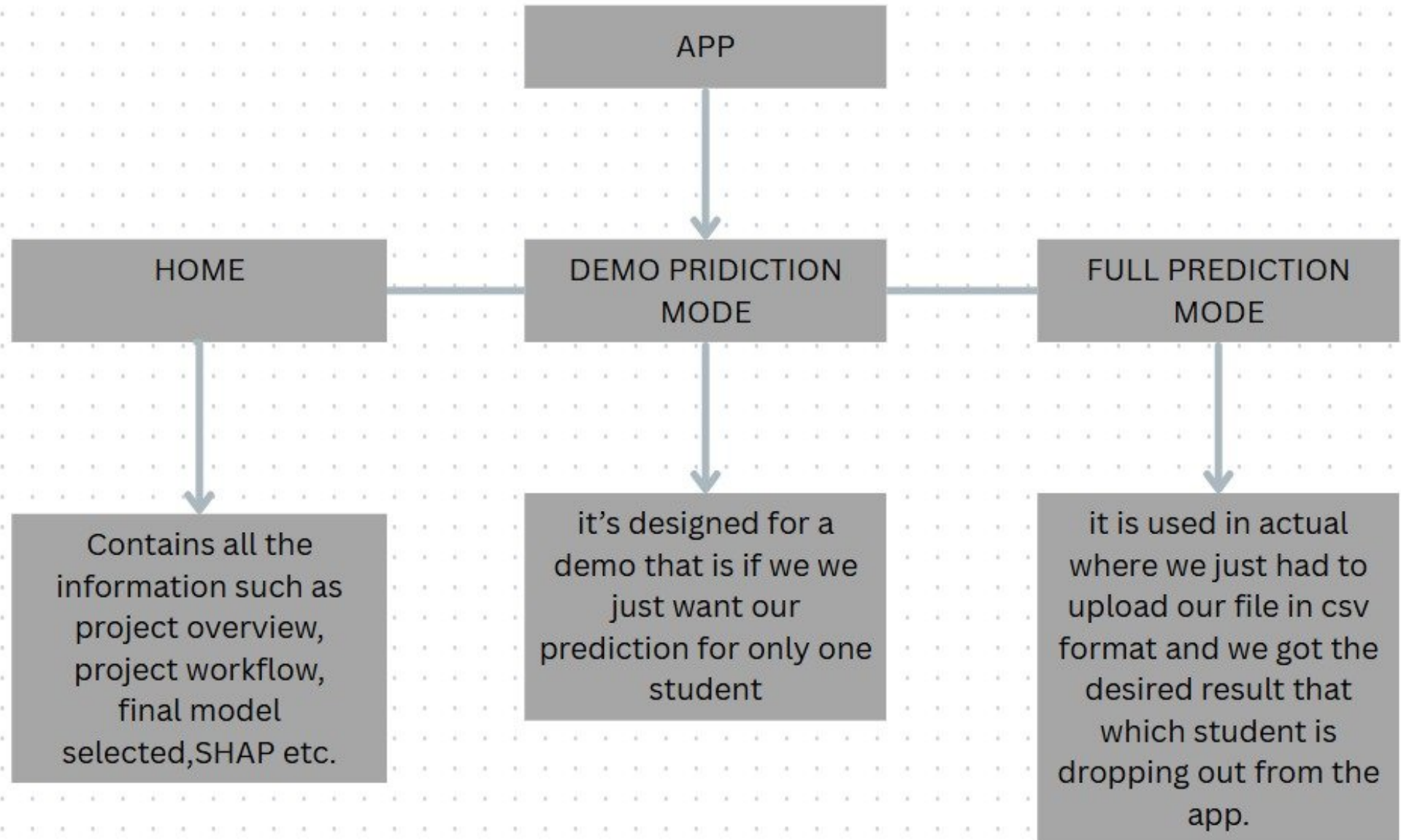


SITE MAP



Strengths and Limitations

Strengths

- Explainable predictions
- Batch prediction support
- Personalized recommendations
- What-if simulation
- SHAP-based interpretation
- Modular software architecture

Limitations

- Trained on OULAD dataset
- Depends on input data quality
- Predictions are probabilistic
- Should not replace human academic judgment
- May require retraining for new institutions

Future Scope

- Real-time LMS integration
- Instructor dashboard
- Early warning notifications
- Intervention tracking
- Continuous model retraining
- Fairness and bias monitoring

Model Selection

Model	Role
Logistic Regression	Baseline Model
Decision Tree	Compared Model
Random Forest	Compared Model
XGBoost	Final Selected Model

Why XGBoost?

XGBoost was selected because it achieved the strongest overall performance and provided the best balance between precision, recall, F1-score, and ROC-AUC.

It also integrates effectively with SHAP, which makes it suitable for explainable student-level predictions.

Final Model Performance

86.90%	73.79%	89.72%	80.98%	94.58%
Accuracy	Precision	Recall	F1 Score	ROC-AUC

Feature Engineering

Data Preparation

Missing Value Handling
Feature Engineering
Assessment Statistics
Learning Engagement Features
Binary Encoding
One-Hot Encoding
Feature Alignment for Deployment

Engineered Features

Average Assessment Score
Maximum Assessment Score
Minimum Assessment Score
Total Assessments
Average Assessment Weight
Total Clicks
Active Learning Days
Last Activity Day

Dataset Information

32,548 ⇄

Student Records

46

Raw Features

76

Encoded Model Features

Dropout

Target Variable

The project uses the Open University Learning Analytics Dataset after cleaning, feature engineering, and transformation into a model-ready format. The final model input contains academic, demographic, assessment, and learning engagement features.

Demo Prediction Mode

Predict student dropout risk using selected meaningful features. This mode is intended for demonstration and educational purposes.

Student Information

Gender

Male



Highest Education

No Formal quals



Age Band

0-35



Disability

No



Previous Attempts

0



Studied Credits

Prediction Result

Prediction results will appear here after clicking **Generate Prediction**.

Learning Engagement

Total Clicks

500

- +

Active Days

60

- +

Last Activity Day

120

- +

Forum Clicks

100

- +

Homepage Clicks

100

- +

Content Clicks

100

- +

Generate Prediction

Academic Performance

Average Score



Maximum Score



Minimum Score



Total Assessments

5

- +

Average Assessment Weight



Full Feature Prediction Mode

Upload the processed final dataset. The application will encode, align features, and generate dropout predictions for all students.

Upload final_dataset.csv

 Upload

200MB per file • CSV

Upload final_dataset.csv to generate predictions for multiple students.

we just have to upload a file into this